

HSBS2001 MATHEMATICS-III (3-0-0)

Module 1: Laplace Transforms (8 Hours)

Laplace transforms, inverse transforms, linearity, shifting, transforms of derivatives and integrals, solution of ODEs, unit step function, Dirac's delta function, differentiation and integration of transforms, convolution, integral equations.

Module 2: Fourier series & Applied PDE's (8 Hours)

Fourier series: Euler's formula, 2π and arbitrary periodic functions, even and odd functions.

Elementary PDE's: Method of separation of variables (simple problems). One dimensional wave equation: solution by separation of variables, One dimensional heat equation: solution by Fourier series.

Module 3: Basic Probability (8 Hours)

Axiomatic definition of probability, Basic properties, conditioning and independence, Random variables (discrete and continuous), probability mass and density functions, cumulative distribution functions, moments of random variables, mean and variance.

Module 4: Probability Distributions (8 Hours)

Discrete Probability distributions: Binomial, Poisson and hyper-geometric distributions. Continuous Probability distributions: exponential, uniform and normal distributions.

Module 5: Applied Statistics (8 Hours)

Random sampling, estimation of parameters, maximum likelihood estimation, confidence intervals. Regression and correlation analysis: fitting of straight lines (method of least squares), correlation coefficient with basic properties.

Text Books:

1. Advanced Engineering Mathematics by E. Kreyszig, John Wiley & Sons Inc. 10th Edition.
2. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers & Keying Ye, "Probability & Statistics for Engineers & Scientists", Eighth Edition, 2007, Pearson Education Inc., New Delhi.

Reference Books:

1. Ordinary and Partial Differential equations by J. Sinha Roy and S. Padhy, Kalyani Publishers.
2. Higher Engineering Mathematics by B. V. Ramana, McGraw Hill Education.
3. Engineering Mathematics by Pal and S. Bhunia, Oxford Publication.
Stochastic Processes, 2nd Edition by Roy D. Yates, Rutgers and David J. Goodman, John Wiley and Sons, INC.

EEPC2001 ELECTRICAL CIRCUIT ANALYSIS (3-0-0)

Module I

8Hours

Graph Theory: Graph of network, Trees, Cotrees and Loops, Number of possible trees of a graph, Incidence Matrix, Cut-set matrix, Tie-set and loop currents, Inter-relationship among various matrices. Concept of duality and dual networks.

Network Theorems (for both AC and DC networks): Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Reciprocity theorem, Compensation theorem, Millman's Theorem, Substitution Theorem.

Module II

9Hours

Time Domain Analysis of First and Second-order networks: Source-free RC circuit, Source-free RL circuit, Singularity Functions, Step Response of an RC Circuit, Step Response of an RL Circuit, Initial and final conditions in network elements, The Source-Free Series RLC Circuit, The Source-Free Parallel RLC Circuit, Step Response of a Series RLC Circuit, Step Response of a Parallel RLC Circuit.

Coupled Circuit Analysis: DOT convention, coefficient of coupling, series and parallel coupled circuits, electrical equivalent of magnetically coupled circuits

Module III

6 Hours

Laplace Transform: Definition of Laplace Transform, properties of Laplace transform, Inverse Laplace transform: (i) simple poles (ii) repeated poles (iii) complex poles, Gate Function, Impulse Function, Laplace Transform of periodic functions, Convolution Integral.

Electrical Circuit Analysis Using Laplace Transform: Representation of Circuit Elements in s-domain, Circuit analysis in s-domain: With and without initial conditions.

Module IV

8Hours

Resonance: Resonance in series and parallel RLC circuit, variation of current and voltage with frequency, selectivity and bandwidth, Q-factor.

Two Port Network Theory: Introduction, Characterization of Linear Time-invariant two-port networks, impedance parameters, admittance parameters, transmission and hybrid parameters, inter-relationship between the parameters, interconnections of two-port networks, Two-port symmetry.

Filters: Introduction to first-order and second-order passive and active filters.

Module V

5Hours

Network Functions: Network functions, poles and zeroes, necessary condition for Driving point function, necessary condition for transfer function, Time domain behaviour from pole zero plot.

Passive network synthesis: Positive real function, Driving point and transfer impedance function, Cauer-I, Cauer-II, Foster-I and Foster-II forms, Driving point and transfer impedance function

Course Outcomes : This course will enable students to:

CO1: apply network theorems for the analysis of electrical circuits

CO2: analyse the transient and steady-state response of electrical circuits.

CO3: apply Laplace Transform for the analysis of electrical circuits.

CO4: analyse the behaviour of two-port networks and synthesis of passive two-port networks.

Text Book(s):

1. Charles K. Alexander, Matthew N. O. Sadiku, "Fundamentals of Electric Circuits" | 7th Edition, McGraw Hill Publication.
2. D. Roy Choudhury, "Networks and Systems", New Age International Publications, 1998.

Reference Book(s):

1. W. H. Hayt and J. E. Kemmerly, "Engineering Circuit Analysis", McGraw Hill Education, 2013.
2. M. E. Van Valkenburg, "Network Analysis", Prentice Hall, 2006.

EEPC2002 Electrical Machines-I (3-0-0)

Module I

6Hours

Electromechanical Energy Conversion: Introduction to magnetic circuits, Magnetically induced EMF and MMF, Review of Faraday's Law and Lenz's Law, Hysteresis and Eddy-current losses.

Module II

8 Hours

DC Generators: Principle of operation, Commutator action, Constructional features, Armature windings, Lap and wave windings, Simplex windings, E.M.F. Equation, Methods of excitation: separately excited and self-excited generators, Build-up of E.M.F., Critical field resistance and critical speed, Armature reaction: Crossmagnetizing and demagnetizing AT/pole, Compensating winding, Commutation, Reactance voltage, Methods of improving commutation, Internal and External characteristics for self and separately excited DC generator, Load characteristics of shunt, series and compound generators, Parallel operation of DC generators.

Module III

8Hours

DC Motors: Principle of operation, Back E.M.F., Torque equation, Characteristics and application of shunt, series and compound motors, Starting of DC motor, Necessity of a starter, Constant & Variable losses, Calculation of efficiency, Condition for maximum efficiency. Speed control of DC Motors: Armature voltage and field flux control methods, Ward Leonard method. Methods of Testing: Brake test, Swinburne's test.

Module IV

8Hours

Single-phase Transformers: Single phase transformer, Constructional details, Principle of operation, EMF equation, Magnetizing current and core losses, Phasor diagrams at no-load and load conditions of an ideal transformer and practical transformer, Equivalent circuit, losses and efficiency, All day efficiency, Voltage regulation, Determination of parameters from tests (Polarity Test, Open Circuit Test and Short Circuit Test, Back to Back test), Parallel operation of transformers and load sharing. Auto Transformer: Basic constructional features, VA conducted magnetically and electrically, Saving of copper, Applications.

Module V

10Hours

Three phase Transformer: Constructional features of three phase transformers, As a single unit and as a bank of three single phase transformers, Three phase connection of transformers: Various phase displacements (0° , 180° , $+30^\circ$ and -30°), Connection diagrams and Phasor diagrams of various vector groups (Yy0, Dd0, Dz0, Yy6, Dd6, Dz6, Yd1, Dy1, Yz1, Yd11, Dy11, and Yz11), Scott connection, Open delta connection, Parallel operation.

Course Outcomes: On completion of this course, students are able to:

CO-1: Understand the concepts of Magnetic circuits.

CO-2: Understand the operation of DC Generators.

CO-3: Understand the operation and concepts of DC Motors.

CO-4: Understand the operation of single-phase transformers & auto transformer.

CO-5: Understand three phase transformers circuits.

Text Book(s):

1. I. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill Education, 2010.
2. P. S. Bimbhra, "Electrical Machinery", Khanna Publishers, 2011.

Reference Book(s):

1. Electric Machinery – Fitzgerald, Charles Kingsley Jr., S. D. Umans – Tata McGraw Hill.
2. Electric Machinery and Transformers – Guru & Hiziroglu – Oxford University Press, 2010.

Syllabus Preparation as per NEP

1. Each subject should have pre-requisites (if any), 5 Modules with minimum 30 contact hours.
2. Each subject COs, POs & PSOs are to be articulated as per Blooms Taxonomy.
3. In each subject, at least two Textbooks and relevant Reference books are to be mentioned.

EOPC2002 ANALOG AND DIGITAL ELECTRONICS CIRCUIT (3-0-0)

Course objectives: This course will enable students to

- Understand Bipolar Junction Transistors and Metal Oxide Semiconductors.
- Analysis of DC biasing of Semiconductor Circuits using BJT and MOSFET.
- Understand Input and Output characteristics of Single Stage Amplifier (both BJT and MOSFET).
- Apprehend characteristics of Feedback and Power amplifier.
- Introduce the concept of digital and binary systems
- inculcate concepts of K-MAP to simplify a Boolean expression
- Facilitate students in designing combinational and sequential logic circuits.

MODULE 1:

[8hrs]

Biasing of BJT: DC Analysis, DC Load line, Operating Point, Fixed bias, Emitter bias, Voltage-divider bias, DC bias with voltage feedback, Bias stabilization.

Small Signals Modelling of BJT and their analysis: The r_e transistor model, Hybrid equivalent model, small signal analysis of CE, CC, CB amplifier. Emitter Follower; Cascade Amplifier, Darlington connections and Current Mirror Circuit.

MODULE 2:

[8 hrs]

Biasing of FET and MOSFET: Fixed bias configuration, Self-bias configuration, Voltage divider bias and design.

Small Signal operation and models of FET and MOSFETs: Small signal equivalent models, Single-stage MOSFET Amplifiers: Common-Source (CS) amplifiers, Common-Source amplifiers with a source resistance, Common-Gate (CG) amplifiers, Common-Drain (CD) or Source follower amplifiers and cascaded system.

MODULE 3:

[4 hrs]

Oscillators and power amplifiers: Positive feedback circuit as Oscillator, Barkhausen's criteria for oscillation, Oscillators (Wien Bridge Oscillator, R-C phase shift oscillator and Crystal Oscillator).

Classification of Power Amplifiers, Power dissipation and power conversion efficiency of Class A, Class B amplifiers, Push-pull amplifier.

MODULE 4:

[5 hrs]

Minimization of Boolean Functions: Canonical logic forms, sum of product & product of sums, Karnaugh maps (two, three and four variable), Don't-care Conditions, Quine-McCluskey technique.

Combinational Circuits: Binary multiplier, Magnitude Comparator, decoder, encoder, priority encoder, Multiplexers, De-multiplexers, Parity generators and Checkers, Code converters.

MODULE 5:

[5 hrs]

Sequential circuits: Latches and Flip-flops (SR, D, JK, T), Master-slave flip-flop, flip-flop conversions, Design and analysis of synchronous binary counter and ripple counters.

Registers: Types of Registers, Serial In - Serial Out, Serial In - Parallel out, Parallel In - Serial Out, Parallel In - Parallel Out.

COURSE OUTCOME: After completion of course, student should be able to

CO1: Understand the characteristics and configurations of single stage BJT and MOSFET amplifiers.

CO2: Design amplifier circuits using BJT, FET and study the low and high frequency response of BJT, FET amplifiers.

CO3: Analyse various power amplifiers and to gain knowledge on various oscillator circuits.

CO4: Understand various types of number systems and their conversions

CO5: Identify the importance of canonical forms in the minimization of Boolean functions in digital circuits.

CO6: Design and implement variety of logical devices using combinational circuits and Sequential circuits.

TEXT BOOKS

1. Microelectronic Circuits – Sedra& Smith, International Student Edition, 5th edition
2. Electronic Devices and Circuit Theory – Robert L.Boylestad and LowisNashelsky, Pearson education, New Delhi 10th edition
3. Digital Logic and Computer Design- M Morris Mano, 10th Edition, Pearson, 2008.

REFERENCE BOOKS

1. Millman's Integrated Electronics –Jacob Millman and Christos Halkias, Chetan D Parikh, Mcgraw Hill
2. Electronic Devices – Floyd, Pearson Education
3. Digital Fundamentals (8th Edition)-Floyd and Jain, Pearson Education Limited.

EEPC2201 ELECTRICAL CIRCUIT ANALYSIS LAB (0-0-3)

Sl. No	Name of the Experiments (Any 08 Experiments)	Hrs.
1.	Validation of Network Theorems using AC circuits (Superposition, Thevenin, Norton, Maximum power transfer)	3
2.	Study of DC and AC transients for R-L, R-C & R-L-C circuits using digital storage oscilloscope	3
3.	Determination of two port network parameters (open circuit and short circuit parameters)	3
4.	Determination of two port network parameters(hybrid and transmission parameters).	3
5.	Frequency response of low pass and high pass filters.	3
6.	Frequency response of band pass and band elimination filters.	3
7.	Determination of self-inductance, mutual-inductance and coupling coefficient of a single-phase two winding transformer representing a coupled circuit.	3
8.	Study of series and parallel connected magnetically coupled circuits	3
9.	Study of resonance in R-L-C series circuit using oscilloscope.	3
10.	Study of resonance in R-L-C parallel circuit using oscilloscope.	3

Course Outcomes: This course will enable students to:

CO1:validate Superposition, Thevenin, Norton, and Maximum Power Transfer theorems in AC networks.

CO2:interpret the transient responses of R-L, R-C, and R-L-C circuits using a digital storage oscilloscope.

CO3:characterize a given two port network using open circuit, short circuit, hybrid and transmission parameters.

CO4:interpret the frequency response plot of low pass, high pass, band pass, and band elimination filters.

CO5:validate the self-inductance, mutual inductance, and coupling coefficient of a single-phase two winding transformer.

CO6:interpret the response of resonance in R-L-C series and parallel circuits using an oscilloscope

All the above experiments are also to be performed in simulation environment preferably using open-source softwares(MULTISIM, PSIM, PSPICE, LTSPICE etc) and through virtual lab platforms

EEPC2202 Electrical Machine-I Laboratory (0-0-3)

Sl. No	Name of the Experiment	Hrs.
1.	Load characteristics of DC shunt/compound generator.	3
2.	Load characteristics of DC series motor.	3
3.	Speed control of DC shunt motor by armature voltage control and flux control method.	3
4.	Determination of critical resistance and critical speed from no load test of a DC shunt generator.	3
5.	Back-to-Back testing of DC machines (Hopkinson's method)	3
6.	Determination of efficiency and voltage regulation by open circuit and short circuit test on single-phase transformer.	3
7.	Parallel operation of two single phase transformers.	3
8.	Study of Open delta and Scott connection of two single phase transformers.	3
9.	To connect and measure the line voltage & phase voltage of three single-phase transformer in (a) star-star (Y-Y) (b) star-delta (Y- Δ) (c) delta-delta (Δ - Δ) (d) delta-star (Δ -Y)	3
10.	Back-to Back test on two single phase transformers.	3

CO1: Understand load characteristics of DC shunt and compound generators and series motors.

CO2: Learn speed control methods for DC shunt motors, including armature voltage and flux control.

CO3: Analyse critical resistance and critical speed determination from DC shunt generator no-load tests.

CO4: Explore back-to-back testing of DC machines using Hopkinson's method.

CO5: Determine efficiency and voltage regulation through open circuit and short circuit tests on single-phase transformers, and study parallel operation and different connection configurations of single-phase transformers.

EOPC2202 ANALOG & DIGITAL ELECTRONICS CIRCUIT LABORATORY (0-0-3)

Course Objective:

1. To understand the operation of the various bias circuits of BJT, JFET and MOSFET.
2. To study the frequency response of BJT, JFET and MOSFET-amplifier.
3. To analyse RC phase shift oscillator and square wave testing of amplifier.
4. To analyse combinational Logic circuits using Karnaugh Maps or Quine McClusky Techniques.
5. To study Decoders, Encoders, Digital multiplexers, Binary comparators, and Flip-Flops.
6. To design and analyse Synchronous and Asynchronous Sequential circuits, registers and Counters.

List of Experiments (At least 10 experiments)

Analog Electronics circuit Experiments (Any Five)

1. Design and simulate voltage divider biasing circuits using BJT-CE configuration and compare the results.
2. Design and simulate of voltage divider biasing circuits using JFEET-CS configuration.
3. Design and simulate of Self-biasing circuits of MOSFET.
4. Determine the frequency response of BJT CE- amplifier: low frequency, Mid-frequency and high frequency response.
5. Determine the frequency response of BJT emitter follower (CC-amplifier) circuit.
6. Determine the frequency response of JFET CS- amplifier Circuit.
7. Determine the frequency response of MOSFET CS- amplifier circuit.
8. Study of RC phase shift oscillator using BJT/OPAMP.
9. Study of Class A, B power amplifier.

Digital Electronics circuit Experiments (Any Five)

1. Design of Binary Multiplier.
2. Design and implementation of code converters gray to binary and BCD to seven segmentdisplay.
3. Design and implementation of a function using Multiplexer (MUX) and De-multiplexer(DEMUX).
4. Design of function using decoder and encoders.
5. Flip-Flop: assemble, test and investigate operations of S-R, D and J-K flip-flops.
6. Shift Registers: Design and investigate the operation of all types of shift registers with Parallel load.
7. Counters: design, assemble and test various ripple and synchronous counters - decimal counter, binary counter with parallel load.

Course Outcome: After completion of the sessional student should be able to

1. Understand about the operation of the various bias circuits of BJT, JFET and MOSFET.
2. Acquire knowledge on the frequency response of different amplifier circuits.
3. Learn the operation of combinational Logic circuits using Karnaugh Maps or Quine McClusky Techniques.
4. Study Decoders, Encoders, Digital multiplexers, Binary comparators, and Flip-Flops.
5. Design and analyse Synchronous and Asynchronous Sequential circuits, registers and Counters.